ficult diseases like different forms of cancer, Fragile X syndrome, AIDS and so on. For example, England-based startup Healx is working on drug delivery for rare diseases through their proprietary AI-based platform called HealNet. The five-year old company—with Dr David Brown, co-inventor of Viagra, as one of its chiefs—has seen success in Fragile X treatment during an ongoing project where the aim was to identify, validate and repurpose approved drugs for the disease, using AI, while reducing production costs.

Fragile X is a genetic disability that leads to various behavioural, developmental and intellectual limitations in a person. So far, no proper medication or cure has been found. With the help of HealNet platform, eight drugs were identified as candidates, and in a span of six months of experimentation on Fragile X-affected mice, three of the drugs showed substantial improvement. Studies were carried out based on four behavioural parameters in mice, namely, open field venturing, nesting, fear conditioning and sociability. One of the three drugs has been selected for further in-depth studies.

In another development last year, Department of Dermatology at Dr D.Y. Patil Medical College, Pune, in collaboration with US-based AI solution provider Polyfins Technology, developed an AI-based platform for treating skin and hair disorders. Outcome of the project is an application called Tibot, which can diagnose 12 high-level dermatological disorders,

Diabetic retinopathy can be analysed using AI

including bacterial or fungal infections, tumours, eczema, alopecia and psoriasis, among others, in addition to 90 secondary conditions.

The machine learning platform uses data based on images of the affected skin area and questions answered by the user on the app, to predict top-three skin conditions the user may be suffering from. It provides confidence score and urgency index for every analysis, enabling the user to prioritise the ailment that requires to be diagnosed and treated quickly. Dr D.Y. Patil Medical College is using the platform in its dermatology section, headed by Dr Sharmila Patil, for trials, and is receiving good results.

More research is on the way that will soon have real-life use cases. Researchers at IBM have come up with a small sensor with a microprocessor that can be attached to the fingertip to predict patients' health deterioration. It has received success in Parkinson's detection so far. The sensor can also be used to measure grip strength or grasping forces, to detect users' emotional and behavioural patterns.

The Industrial Technology Research Institute of Taiwan has developed an AI platform that, based on image analysis and data mining, can detect the stage and complexity of diabetic retinopathy—a disease where long-term diabetic patients are at risk of vision loss. The platform is set to be utilised in hospitals across Taiwan to evaluate the effectiveness of the solution. The team behind the platform plans to include cloud technology in the platform, which will also allow doctors to provide remote consultation.

Opportunities in hand

Medical professionals have identified the scope of improving success rates and quickness in treatments while reducing the mortality rate with the help of AI tools. This opens up opportunities for AI solution providers to find a high demand market. A 2018 report by Frost and Sullivan suggests that AI for healthcare will globally become a US$ 6.16 billion market by 2022, with a CAGR of 68.5 per cent. It will bring about savings of US$ 150 billion for the healthcare industry. However, only 15 per cent of healthcare customers have adopted AI properly, which leaves a massive opportunity of improvement to be filled by healthcare businesses, and a huge demand for AI developers.

India has a strong software base and is growing its embedded hardware capabilities, too, promising a good scope for creating the desired AI-based healthcare ecosystem. How the players seize the day will be an important factor.

—Paromik Chakraborty, technical journalist, EFY

2 SMART INSPECTION TECHNOLOGIES Reduce Industrial Opex

Inspection processes we had in past were unreliable, time-consuming and unsafe for humans. To address these challenges at the industrial scale, smart inspection solutions are taking over

In asset-intensive industries, inspections are essential. Any unplanned shutdown can result in huge losses. In manufacturing firms, construction or giant public sector industries, assets are the major investment and productivity is directly dependable on them. It is important to be aware if machines are underperforming to take timely necessary actions for optimum utilisation of assets.